Xia CHEN

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Research associate

Data Scientist with 4+ years of academic research & industry experience in developing data algorithm solutions, bringing insights from data for optimization purposes in diverse industries. Proficient in predictive modeling, data-based value proposition, machine learning in human-computer interaction with the research focus on causality, integration of knowledge-based method, as well as uncertainty analysis. Passion for collaboration organization, creativity, digitalization, and delivering AI-based adaptive solutions.

Work & Research Experience

2020/07 – Current (Anticipated graduation dates: 01.2024)

PhD candidate in Digital Architecture and Sustainability institute, Technische Universität Berlin, Berlin, Germany

- DFG project (German Research Foundation) FOR 2363: Research in data-driven / informed machine learning framework for decision-making aids, uncertainty analysis, and reasoning.
- Lecturer "Artificial Intelligence for Architecture" at the Institute of Digital Architecture, Technical University Berlin

10/2017 – 12/2019

Research Assistant in FCN institute of E.ON Energy Research Center, Aachen, Germany

- Energy time-series data analysis & research (forecasting, clustering)
- Application development of virtual energy system laboratory project.
- Full-stack development, *Juniorprofessur für Energieressourcen- und Innovationsökonomik* (JERI) https://www.jericho-energy.de/

11/2016 - Current

Co-founder of Start-up www.jiaonengwang.com, Shanghai, PR China

- Data solution & consulting in domain of energy digitalization toward research institute, national grid, and government in China

Social Commitment

08/2020 - Current

CINB (Association of Chinese Engineers for Sustainable Construction e.V.)
Executive Committee Member

06/2016 - Current

CEED (Association Chinese Engineers for Renewable Energy in Germany e.V.)
Executive Committee Member

Education

10/2015 - 12/2018

RWTH Aachen University, Aachen, Germany
Master of Science in Sustainable energy supply technology

09/2014 - 09/2015

Beuth Hochschule für Technik Berlin, Berlin, Germany
Bachelor of Engineering in building engineering technology

09/2011 - 09/2015

Tongji University, Shanghai, PR China
Bachelor of Engineering in building facility intelligence, *Faculty of Chinese-German University of Applied Sciences (CDHAW)*

Achievements & Technical Competency

02/2021 — Top 5% in M5 (Makridakis Competitions) competition

07/2021 — Finalist of Siemens “Hello Future” innovation challenge 2021, Digitally-enabled applications for smart districts

General Data Skills — Time-series: Forecasting, Classification, Decomposition, Abnormal Detection, Clustering, Dimension Reduction; Machine Learning, Visualization Techniques, Uncertainty Analysis, Probabilistic Analysis, Causal Inference, Explainable Artificial Intelligence, Bayesian.

Programming language

- Proficient: Python, R
- Familiar with: SQL, JavaScript, HTML5, Vue, PHP

Publications

- Xia Chen; Philipp Geyer (2022) “Machine assistance in energy-efficient building design: A predictive framework toward dynamic interaction with human decision-making under uncertainty” In : Applied Energy, vol. 307, p. 118240. DOI: 10.1016/j.apenergy.2021.118240.
- Xia Chen; Xiaoye Cai; Alexander Kümpel; Dirk Müller; Philipp Geyer. (2022). “Dynamic Feedforward Strategy Development for Building Heating System based on AI Forecasting and Simulation”, accepted by Passive and Low Energy Architecture, PLEA 2022
- Xia Chen, Johannes Staudt, Manuel Margesin, Werner Lang, Philipp Geyer. (2022). “Data-driven and knowledge-based performance evaluation for machine assistance in building design decision support”, accepted by 29th International Workshop on Intelligent Computing in Engineering, EG-ICE 2022. Aarhus, Denmark. 2021
- Xia Chen, Jimmy Abualdenien, Manav Mahan Singh, André Borrmann, Philipp Geyer. (2022). ”Introducing causal inference in the energy-efficient building design process”, submitted, under review in Applied Energy
- 陈夏，张怡卓，蔡晓烨．欧盟-德国建筑碳中和前沿［Ｊ］．暖通空调，２０２２，５２（３）：２５-３８．
- Xia Chen, Tong Guo and Philipp Geyer. (2021). “A hybrid-model time-series forecasting approach for reducing the building energy performance gap” In 28th International Workshop on Intelligent Computing in Engineering, EG-ICE 2021. Berlin, 2021, special issue on Advanced Engineering Informatics
- Xia Chen, Manav Mahan Singh and Philipp Geyer. (2021). “Component-based machine learning for predicting representative time-series of energy performance in building design” In 28th International Workshop on Intelligent Computing in Engineering, EG-ICE 2021. Berlin, Germany. 2021
- Geyer, Philipp, Manav Mahan Singh, and Xia Chen. "Explainable AI for engineering design: A unified approach of systems engineering and component-based deep learning." arXiv preprint arXiv:2108.13836 (2021).
- Xia Chen, Lars Nolting, Jan Priesmann. “FAST- model: An automated protocol for univariate time series Forecasting Algorithm SelecTion” , To be submitted paper

Berlin, March 02. 2022